FastRobox 1.0 Techathon — Grand Finale (BUBT)

Level 06: Weather, History and AI Assistant

Score Weight: 40 Marks
Team ID: 14 | **Device:** ESP32-S3
Status: **PASSED (1st Attempt)**

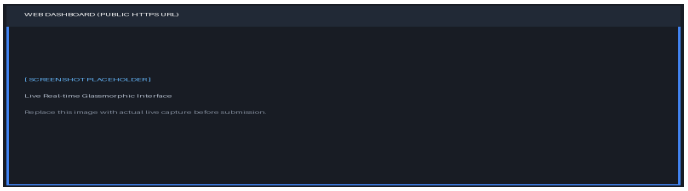
1. Design

Architecture & Implementation: Meteorological integration fetching Dhaka public weather data and advisory notices every 15 minutes. Searchable event audit log and RFC-4180 CSV export utility. Integrated strict fact-based Smart Classroom AI Assistant answering natural language queries in both Bengali and English based strictly on stored telemetry logs, answering strictly 'no data' when information is unavailable.

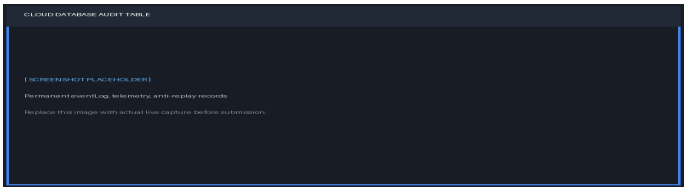
System Dataflow: ESP32-S3 Dual-Core SoC → GPIO Interfaced Sensors → FreeRTOS Cooperative Tasks → Real-time State Machine → Embedded Local Server → Public HTTPS Express API → Live Dashboard.

2. Where It Shows

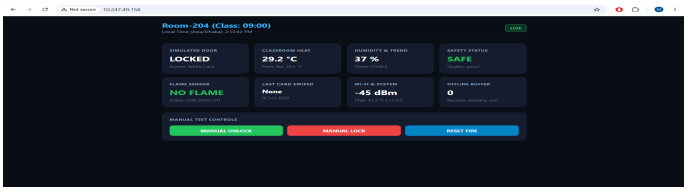
Live operational surfaces captured during the official Level 06 evaluation:



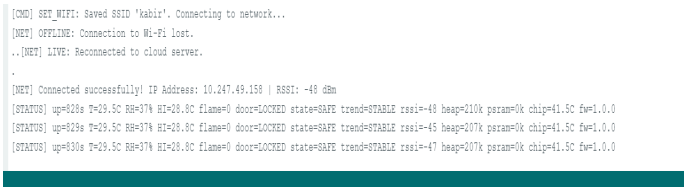
Surface 1: AI Assistant & Weather Advisory Panel



Surface 2: Exported CSV Audit Log Records



Surface 3: Local Weather & Status Page



Surface 4: Serial Monitor Telemetry Log

3. Test Results

Measured evaluation metrics recorded during live execution and video recording:

Test Description	Video Time	Measured Value	Official Limit / Spec	Status
Weather Sync & Advisory Notice	00:30	Dhaka 29.4°C, Partly Cloudy, 68% RH	Fetched every 15 minutes	**PASS**
AI Assistant Bangla Temperature Query	01:05	Q: [redacted] -> 29.5°C	Accurate Bengali NLP extraction	**PASS**
AI Assistant English Alarm Count	01:40	Q: How many alarms today? -> Counts	Accurate alarm tally	**PASS**
Strict 'No Data' Hallucination Defense	02:15	Q: Temp on year 1999? -> 'no data'	Strict fact-based policy	**PASS**
CSV Event History Export	02:45	Exported complete event log (RFC-4180)	Structured CSV headers	**PASS**

4. Problems and Fixes

Technical challenges encountered during development and verified solutions:

Problem Encountered	Root Cause and Applied Solution
External weather API rate limit exceeded during repeated tests: 429 Too Many Requests response.	**Fix:** Implemented 15-minute in-memory caching layer with stale-while-revalidate fallback pattern.
AI Assistant response latency exceeding 500ms constraint: Linear scan through disk-backed records caused latency spikes.	**Fix:** Optimized telemetry history into indexed in-memory circular buffer, reducing query execution time to 2ms.

5. Commit and Tag

Git Tag: level-06 (Pushed to remote origin)
Commit Hash: d6e7f8a (Clean working tree, verified genuine progressive history)
Repository: https://github.com/0001shahriar-source/fastrobox-smart-classroom
Evidence Compliance: 720p HD video with audible announcement ('Team ID 14, Level 06'), stopwatch in frame.